Combinatorics – Notes written by Ike Glassbrook, based on the course and lecture notes of Alex Scott, 2025

Chains, antichains, shadows

Noting that a complete matching of a bipartite graph from X to Y is a vertex disjoint collection of edges such that we cover X.
Theorem 1 (Hall's theorem) *$G = (X \cup Y, E)$ a bipartite graph has a complete matching from X to Y if and only if for all $S \subseteq X$, $|\Gamma(S)| \geq |S|$.*

We then claim that there is a partition of $\mathcal{P}(n)$ into $\binom{n}{\lfloor n/2 \rfloor}$ chains, following as we can find matchings from layer to layer when viewed as Q_n, and there are $\binom{n}{\lfloor n/2 \rfloor}$ elements in the centre allowing us to choose that number of such chains through the layers. Thus we can then prove Sperner's lemma:

Theorem 2 (Sperner's lemma) *An antichain in $\mathcal{P}(n)$ has size at most $\binom{n}{\lfloor n/2 \rfloor}$.*

This is clear as a chain and an antichain meet on at most one element, so partitioning $\mathcal{P}(n)$ into some number of chains means having an antichain with more elements would mean it shares two elements with a chain.

Theorem 3 (LYM Inequality) *Let $\mathcal{F} \subseteq \mathcal{P}(n)$ be an antichain. Then*

$$\sum_{i=0}^{n} \frac{|\mathcal{F} \cap [n]^{(i)}|}{\binom{n}{i}} \leq 1,$$

and we have equality iff $\mathcal{F} = [n]^{(i)}$ for some $i \in [n]$.

Definition 1 *With $\mathcal{F} \subseteq X^{(k)}$ a k-uniform family on X, the lower shadow $\partial\mathcal{F}$ of $\mathcal{F}$ is $\bigcup_{A \in \mathcal{F}} A^{(k-1)}$.*

Essentially, to form a shadow on a k-uniform family $\mathcal{F}$, from each $A \in \mathcal{F}$ we pick $k-1$ elements.

This is proven using a local version with $\mathcal{F} \subseteq [n]^{(r)}$, that

$$\binom{n}{r}|\partial\mathcal{F}| \geq \binom{n}{r-1}|\mathcal{F}|,$$

shown via double counting of edges in Q_n, noting that each element in $\mathcal{F}$ contains k sets of size $k-1$, and each element in $\partial\mathcal{F}$ is contained by $\leq n-(k-1)$ elements in $\mathcal{F}$, so $(n-k+1)|\partial\mathcal{F}| \geq k|\mathcal{F}|$. Equality occurs $\mathcal{F} = [n]^{(k)}$ or $\mathcal{F} = \varnothing$.

As a result, if $\mathcal{F}$ is an antichain in $\mathcal{P}(n)$, $|\mathcal{F}| \leq \binom{n}{\lfloor n/2 \rfloor}$ with equality for $\mathcal{F} = [n]^{(k)}$, $k \in \{\lfloor n/2 \rfloor, \lceil n/2 \rceil\}$.

These results can also all be shown probabilistically. If we take a random set in $\mathcal{P}(n)$, then the LHS of the LYM inequality is just the probability that this random set is in $\mathcal{F}$.

Theorem 4 (Dilworth's theorem) *Let $(P, \leq)$ be a finite poset. The minimum number of chains needed to cover P is the maximum size of an antichain.*

One direction of this is clear: the minimum number of chains needed to cover P is certainly not less than the maximum size of an antichain, because then we would have an antichain of maximum size which contains more than one element in one of the covering chains, giving a contradiction.

The other direction is slightly more difficult, but the key idea is to provide a construction of a chain covering which is of equal size as the largest antichain. Drawing a picture, this looks possible but fiddly for large $|P|$, and thus we opt for strong induction. Our goal is to find a proper decomposition of P that will give us chain coverings by induction, and we hope to use these to construct the covering of P. In particular, we want to decompose P *vertically*, cutting it along an antichain, so that we might then link up the chains after decomposition.

Labelling an antichain of maximum size A, we consider first the case that we can just decompose P by taking a maximal chain C. If $P \setminus C$ contains no antichain of size $|A|$, then it may be covered by $|A|-1$ chains, and thus we are done. Otherwise, we split P into $S^+ = \bigcup_{a \in A}\{x : x \geq a\}$ and $S^- = \bigcup_{a \in A}\{x : x \leq a\}$, and may argue that these are each proper subsets, as the maximal and minimal elements of C evade each of them.

We also note a dual to this theorem:
Theorem 5 (Mirsky's theorem) *Let $(P, \leq)$ be a finite poset. The minimum number of antichains needed to cover P is the maximum size of a chain.*

To prove this, we calculate the height of a maximal chain starting at each element $v \in P$, and note that we can get antichains by selecting the elements with a particular height, for each possible height.

Theorem 6 (Erdös) *With $\boldsymbol{x} \in \mathbb{R}^n$ s.t. $x_i \geq 1$ for all $i \in [n]$. For every $\alpha \in \mathbb{R}$, there are at most $\binom{n}{\lfloor n/2 \rfloor}$ subsets $I \subseteq [n]$ such that $\sum_{i \in I} x_i \in [\alpha, \alpha+1)$.*

By using Sperner's lemma, we see that if more such subsets existed, we would have one containing the other, and thus violating the required property.

Indeed we can extend this theorem to the case $|x_i| \geq 1$, noting that replacing $+x_i$ with $-x_i$ just switches around whether we're making an 'include' or 'don't include' decision, translating the collection of possible sums.

We want to generalise this a bit more, and in order to do this we introduce the concept of chain symmetry – we say that a chain in $\mathcal{P}(n)$ is symmetric if each link adds one element, and if the smallest and largest sets in the chain have gross cardinality n. Indeed we can partition $\mathcal{P}(n)$ into symmetric chains by induction. Critically, as every symmetric chain must contain an element in $[n]^{(\lfloor n/2 \rfloor)}$, thus there are $\binom{n}{\lfloor n/2 \rfloor}$ chains in such a decomposition.

Theorem 7 (General dimension Erdös) *Let $k, n \geq 1$, suppose that $\boldsymbol{X} = [\boldsymbol{x}_1, \dots, \boldsymbol{x}_n] \in \mathbb{R}^{k \times n}$ s.t. $||\boldsymbol{x}_i||_2 \geq 1$ for all i. With $K \subseteq \mathbb{R}^k$ s.t. $\operatorname{diam}(K) < 1$, there are at most $\binom{n}{\lfloor n/2 \rfloor}$ subsets $I \subseteq [n]$ s.t. $\sum_{i \in I} x_i \in K$.*

It is sufficient to show that we can partition $\mathcal{P}(n)$ into $\binom{n}{\lfloor n/2 \rfloor}$ sets $D_1, \dots, D_{\binom{n}{\lfloor n/2 \rfloor}}$ which are 'sparse', meaning that for $A, B \in D_i$, $||\sum_{i \in A} \boldsymbol{x}_i - \sum_{j \in B} \boldsymbol{x}_j|| \geq 1$. If such a partition exists then we must only be able to select one valid set from each collection in the partition, and the result is proven.

To prove this, we find a symmetric partition $\mathcal{P}(n)$ into sparse families by induction. For a symmetric partition $\mathcal{F}_1, \dots, \mathcal{F}_m$, order the elements of $\mathcal{F}_i$ as $A_1, \dots, A_t$ where $\langle x_{A_1}, x_n \rangle \leq \dots \langle x_{A_t}, x_n \rangle$, and we hope to then generate new chains for which the sets more aligned with x_n are at the top (and thus $x_n + x_{A_t}$ is kept far from the other sets). We do this and find straightforwardly that with the same procedure to generate a symmetric partition in the first place subject to this particular reordering, we get the desired result.

Kruskal-Katona

We would like to get a more precise understanding of the relative cardinalities of r-uniform sets and their shadows. We get an immediate inequality via LYM (that the density of a shadow in $[n]^{(r-1)}$ is greater than the density of the original set in $[n]^{(r)}$), but want to understand where we get closer to and further from the equality. To do this, we define lexicographic and colexicographic order:

In *lex*, for $A, B \in [n]^{(r)}$, $A <_{\text{lex}} B$ if $A \neq B$ and $\min(A \triangle B) \in A$. Thus in lex order, the place of a set in an order is primarily determined by its lowest element (and then its next lowest, and on and on). We therefore get in $r = 3$,

$$123,\ 124,\ 125,\ \dots,\ 12(n),\ 134,\ 135,\ \dots, 1(n-1)(n), 234, \dots.$$

To compute the successor of $A = \{a_1, \dots, a_r\}$ written in order, increment the largest i such that $a_i < n + r - i$, and then reset each a_j to its smallest possible value for $j > i$.

In *colex*, for $A, B \in [n]^{(r)}$, $A <_{\text{colex}} B$ if $A \neq B$ and $\max(A \triangle B) \in B$. Thus in colex order, the place of a set in an order is primarily determined by its highest element (and then its next highest, and on and on). Thus in $r = 3$,

$$123,\ 124,\ 134,\ 234,\ 125,\ 135,\ 235,\ 145,\ 245,\ 345,\ \dots.$$

Crucially, we get a concentration of the same numbers in a way not seen in lex, ensuring that the possible shadows created in any prefix in colex order ends up being small. Note that for a particular set $A = \{a_1, \dots, a_r\}$ written in order, we can compute the successor of A by incrementing the first a_i such that $a_i + 1 < a_{i+1}$, and resetting all a_j to j for $j < i$.

Colex may be said to be dual to lex. If we write $\text{lex}, \text{colex} : [n]^{(r)} \to [\binom{n}{r}]$ for the index of each element under each ordering, we see that

$$\left(x \mapsto \binom{n}{r} + 1 - x\right) \circ \text{colex} = \text{lex} \circ (x \mapsto n + 1 - x)$$

To see this, note that $A <_{\text{colex}} B$ iff $A \neq B$ and $\max(A \triangle B) \in B$, which is equivalent to $\min((n+1-A) \triangle (n+1-B)) \in n+1-B$, and this in itself equivalent to $n+1-A >_{\text{lex}} n+1-B$. Thus colex order is equivalent to reversed lex order on the reversed order on $[n]$, giving us our duality relation. If we have intuitively that colex gives a concentrated ordering while lex gives a sparse ordering, then this explains it: if we don't care about the underlying symbol set, lex is just reversed colex, and thus as sparse as colex is concentrated.

Helpfully, we see that the weight of a particular element in colex can be quantified straightforwardly:

$$w_{\text{colex}}(A) = \sum_{i \in A} 2^i.$$

The idea here being that each consecutively 2^i term will dominate over any sum of 2^js for $j < i$, so $w_{\text{colex}}(A) < w_{\text{colex}}(B)$ iff the largest i on which the two sets differ occurs in B. Using duality, we can get the slightly less natural, but still valid expression

$$w_{\text{lex}}(A) = 1 - \sum_{i \in A} 2^{-i}.$$

In this case it is the low-value terms which dominate the sum, pushing it towards the front.

Kruskal-Katona gives us a language for this notion of 'concentratedness'.

Theorem 8 (Kruskal-Katona) *Let $\mathcal{F} \subseteq [n]^{(r)}$, $\mathcal{A}$ be the family of the first $|\mathcal{F}|$ elements of $[n]^{(r)}$ in colex order. Then $|\partial\mathcal{F}| \geq |\partial\mathcal{A}|$.*

Essentially, the claim is that colex order creates the most overlap in the shadow, thus minimising its size. To show this takes some time, but fundamentally we do this by reducing $\mathcal{F}$ to a family which is an initial segment of colex, and demonstrate that at each step $|\partial\mathcal{F}'| \leq |\partial\mathcal{F}|$. Intuitively, our goal is to swap large elements for small elements, to get ourselves towards colex.

We define a compression operator to swap in i instead of j and j instead of i wherever this is coherent, and to leave all other sets in $\mathcal{F}$ untouched otherwise. We refer to this as a left compression where $i < j$, and conjecture that left compressions should not increase the shadow.

$$C_{ij}(A) = \begin{cases} (A \setminus j) \cup i & \text{if } i \notin A, j \in A \\ A & \text{otherwise} \end{cases}$$
$$C_{ij}(\mathcal{F}) = \{C_{ij}(A) : A \in \mathcal{F}\} \cup \{A \in \mathcal{F} : C_{ij}(A) \in \mathcal{F}\}.$$

Indeed, the shadow of $C_{ij}(\mathcal{F})$ is no larger than the shadow of $\mathcal{F}$, and there is a family $\mathcal{A}$ which is a fixed point of all left-compressions (which we call left-compressed). Unfortunately, not every left-compressed family is in colex form, as $\{12, 13, 14\}$ is left-compressed but contains 14, which we can't get rid of e.g. via C_{34} because we get it back from 13. Indeed we want to be able to convert this into $\{12, 13, 23\}$, but this is impossible without doing two steps simultaneously.

Thus we want to replace i and j with disjoint sets $U, V \in \mathcal{P}(n)$, $|U| = |V|$, and technical details allow us to get the compression as desired.

We write

$$C_{UV}(A) := \begin{cases} (A \setminus V) \cup U & \text{if } U \cap A = \varnothing, V \subseteq A \\ A & \text{otherwise.} \end{cases}$$
$$C_{UV}(\mathcal{F}) := \{C_{UV}(A) : A \in \mathcal{F}, C_{UV}(A) \notin \mathcal{F}\} \cup \{A : A \in \mathcal{F}, C_{UV}(A) \in \mathcal{F}\}.$$

Note that (in the sense of the operator defined on $\mathcal{P}(n)$) $C_{UV} \circ C_{UV} = C_{UV}$, making this description equivalent to the one presented in the lecture notes. We see that this is essentially a 'replace V with U whenever this is clean' operator, and so the operator on $\mathcal{P}(\mathcal{P}(n))$ is defined to replace each set by the swapped set whenever it creates something new.

Our concern is specifically how this deals with shadows. We claim in particular that if for any $u \in U$, there is $v \in V$ such that $\mathcal{F}$ is $(U \setminus u, V \setminus v)$ compressed, then $|\partial C_{UV}(\mathcal{F})| \leq |\partial\mathcal{F}|$, with the idea being that $C_{VU}(\partial C_{UV}(\mathcal{F}) \setminus \partial\mathcal{F}) = \partial\mathcal{F} \setminus \partial C_{UV}(\mathcal{F})$, and is injective.

Once this is clear, we argue that if $\mathcal{A}$ is of minimal colex weight subject to $|\mathcal{A}| = |\mathcal{F}|$, $|\partial\mathcal{A}| \leq |\partial\mathcal{F}|$, then $\mathcal{A}$ is an initial segment of colex. To do this, look at its compression via

$$\Lambda = \{(U, V) : |U| = |V| > 0, U \cap V = \varnothing, \max U < \max V\}.$$

Either it is already Λ compressed in which case it can be seen that it is an initial segment of colex, and if it is not then using the above lemma applied to a minimal size U (which can only exist if the set of (U, V) for which $\mathcal{A}$ is not compressed is non-empty), we get a contradiction.

Intersections

We say that $\mathcal{A} \subseteq \mathcal{P}(n)$ is intersecting if $A \cap B \neq 0$ for $A, B \in \mathcal{A}$. It's immediately clear that $|\mathcal{A}| \leq 2^{n-1}$.

Theorem 9 (Erdös-Ko-Rado) *For $r \leq n/2$, $\mathcal{A} \subseteq [n]^{(r)}$ intersecting implies that $|\mathcal{A}| \leq \binom{n-1}{r-1}$.*

There are two separate proofs for this: the first involves Katona's circle method, and the second using the Kruskal-Katona theorem. The circle method idea is to double count, by counting the number of edges in the corresponding bipartite graph, the number of pairs (f, A) where $f : [n] \to [n]$ is a bijection, $A \in \mathcal{A}$, and $f(A)$ is a contiguous sequence modulo n (so allowing $n \to 1$ as a step in the interval). Double counting allows us to get an upper bound on the number of As for each f, and a lower bound on the number of fs for each A, giving the desired result.

The Kruskal-Katona idea is to instead think in terms of shadows – noting that any shadow (and indeed repeated shadow) of $\{A^c \in [n]^{(n-r)} : A \in \mathcal{A}\}$ is disjoint from all sets in $\mathcal{A}$, and thus we can use Kruskal-Katona to get a lower bound on the size of these shadows.

Application of EKR

Theorem 10 (Liggett's theorem) *With $\boldsymbol{Y} \overset{\text{i.i.d.}}{\sim} \operatorname{Ber}(p)^n$, $\boldsymbol{\alpha} \geq 0$, $\sum \alpha_i = 1$,*

$$\mathbb{P}\left(\sum_i \alpha_i Y_i \geq 1/2\right) \geq p.$$

This theorem is proven only in the case where for no $I \subseteq [n]$ do we have $\sum_{i \in I} \alpha_i = 1/2$. The idea will be to track $\sum Y_i \sim \operatorname{Bin}(n, p)$, and deterministically calculate the number of 'satisfying' instances for each value of $\sum Y_i$. The key facts are then that with $\mathcal{A}_k = \{A \in [n]^{(k)} : \sum_{i \in A} \alpha_i > 1/2\}$, $|\mathcal{A}_k| + |\mathcal{A}_{n-k}| = \binom{n}{k}$ by taking the complement, and that $\mathcal{A}_k$ must be intersecting for each k. We then want to apply EKR to $\mathcal{A}_{n-k}$ for $k \geq n/2$, thus demonstrating that many instances are satisfactory for large k, and that this overcomes the issues for low k.

Theorem 11 (Two Families Theorem) *Let $A_1, \dots, A_k$, $B_1, \dots, B_k$ be finite sets such that each (A_i, B_i) pair is disjoint, but no other (A_i, B_j) pair is disjoint. Then*

$$\sum_{i=1}^{k} \binom{|A_i| + |B_i|}{|A_i|}^{-1} \leq 1.$$

Observe immediately that each summand is equal to the probability that we choose the (A_i, B_i) partition of $A_i \cup B_i$. In this case, the joint distribution on our choices for each i is specified by selecting, uniformly, an ordering on the elements of all sets, and then determining the probability that all elements of each A_i occur before their corresponding B_i.

Theorem 12 (Kleitman's theorem / Harris lemma) *Let $\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}(n)$ be down-sets. Then*

$$|\mathcal{A} \cap \mathcal{B}| \geq 2^{-n}|\mathcal{A}||\mathcal{B}|$$

This may be proven by standard techniques with induction on n. The intuition for this should be that $|\mathcal{A} \cap \mathcal{B}|/2^n \geq |\mathcal{A}|/2^n \cdot |\mathcal{B}|/2^n$, and thus that down-sets must intersect considerably (in particular, that if we include each element in $[n]$ with probability $1/2$, $\mu(A \cap B) \geq \mu(A)\mu(B)$). This has an important application in problem sheet 3, and more generally has applications wherever we want to bound the cardinality of families which are likely to agree considerably due to intersection properties.

Note that this comes with an immediate (unless you're me, panicking 5 minutes into a probabilistic combinatorics paper, apparently) corollary for up-sets. For any down-set $\mathcal{A}$, assume that $C \supseteq B \notin \mathcal{A}$. Then if $C \in \mathcal{A}$, thus $B \in \mathcal{A}$ which is a contradiction, so clearly $\mathcal{A}^c$ is an up-set. Therefore, for $\mathcal{A}$ a down-set, $\mathcal{C}$ an up-set,

$$\begin{aligned} |\mathcal{A} \cap \mathcal{C}| &= |\mathcal{A}| - |\mathcal{A} \cap \mathcal{C}^c| \\ &\leq |\mathcal{A}| - 2^{-n}|\mathcal{A}||\mathcal{C}^c| \\ &= |\mathcal{A}|(1 - n2^{-n}) + 2^{-n}|\mathcal{A}||\mathcal{C}| \\ &\leq 2^{-n}|\mathcal{A}||\mathcal{C}| \end{aligned}$$

where we can see the case separately for $n = 1$.

Definition 2 *$\mathcal{A} \subseteq \mathcal{P}(n)$ is t-intersecting if $|A \cap B| \geq t$ for all $A, B \in \mathcal{A}$.*

Theorem 13 (Generalised Erdös-Ko-Rado) *With $1 < t \leq k$ integers, there is an integer n_0 such that for $n > n_0$, $\mathcal{A} \subseteq [n]^{(k)}$ is t-intersecting implies*

$$|\mathcal{A}| \leq \binom{n-t}{k-t}$$

with equality iff $\mathcal{A} = \{A \in [n]^{(k)} : T \subseteq A\}$ for some $T \in [n]^{(t)}$.

This isn't *quite* a full generalisation due to the "for large enough n" condition, but its proof is comparatively simple. The point of this is to give us even stronger bounds on sets which we know are 'very intersecting'.

The argument is a case distinction with respect to $A, B \in \mathcal{A}$ for which $|A \cap B| = t$.
Sunflower: for all $C \in \mathcal{A}$, $A \cap B \subseteq C$, and so the number of possible sets is exactly $\binom{n-t}{k-t}$.
Semi-sunflower: there exists $C \in \mathcal{A}$ such that $A \cap B \not\subseteq C$. Thus every $D \in \mathcal{A}$ must intersect $A \cup B \cup C$ on $\geq t+1$ elements (otherwise it fails to be t-intersecting on A, B, or C). Thus there exists at most $\binom{|A \cup B \cup C|}{t+1}\binom{n-t-1}{k-t-1}$ sets satisfying this property in $\mathcal{A}$, and for n large enough

$$\begin{aligned} \binom{|A \cup B \cup C|}{t+1}\binom{n-t-1}{k-t-1} &\leq |A \cup B \cup C|^{t+1}(n-t-1)^{k-t-1} \\ &\leq (3k)^{t+1}n^{k-t-1} \\ &< \left(\frac{e(n-t)}{k-t}\right)^{k-t} \\ &\leq \binom{n-t}{k-t} \end{aligned}$$

where the third inequality holds for n large enough by standard polynomial asymptotics.

The equality result can be seen immediately, as it may hold only in the sunflower case.

It turns out we get a distinctly tight bound when the sense of t-intersection is strict:
Theorem 14 (Fisher's inequality) *Let $k \geq 1$. If $\mathcal{A} \subseteq \mathcal{P}(n)$ satisfies $|A \cap B| = k$ for all distinct $A, B \in \mathcal{A}$, then $|\mathcal{A}| \leq n$.*

This follows from a new type of argument which we continue to use in this section: to identify sets with their characteristic function (a vector in $\mathbb{F}_2^n$), and then use properties of linear independent sets. In this case, we just argue that if $\mathcal{A} \subseteq [n]^{(>k)}$, then $\{\chi_A : A \in \mathcal{A}\}$ is linearly independent, which is obtained by looking at the inner product.

Theorem 15 (Oddtown theorem) *Let $\mathcal{A} \subseteq \mathcal{P}(n)$ be a family such that $|A|$ is odd for $A \in \mathcal{A}$ and $|A \cap B|$ is even for all distinct $A, B \in \mathcal{A}$. Then $|\mathcal{A}| \leq n$.*

This is done using exactly the same argument as above.

Theorem 16 (Modular Frankl-Wilson theorem) *Let p be prime, $S \subseteq \{0, 1, \dots, p-1\}$, and $\mathcal{A} \subseteq \mathcal{P}(n)$ satisfies*
- *$|A| \notin S \bmod p$ for all $A \in \mathcal{A}$, and*
- *$|A \cap B| \in S \bmod p$ for all $A, B \in \mathcal{A}$ distinct.*

Then

$$|\mathcal{A}| \leq \sum_{i=0}^{|S|}\binom{n}{i}.$$

The proof of this involves once again working in $\mathbb{F}_p^n$. For $A \in \mathcal{A}$, define

$$f_A(x) = \prod_{s \in S}\left(\sum_{i \in A} x_i - s\right)$$

Now on the basis that we only want to evaluate our polynomials on characteristic vectors, we can 'multilinearise' them, replacing each power x^k by x, and restricting to $\mathbb{F}_2$, in order to squeeze down the dimension of the space within which we're working. We then get a set of linearly independent multilinear polynomials of degree at most $|S|$, and as this is spanned by the monomials $\{\prod_{i \in A} x_i : |A| \leq |S|\}$, the result follows by the same principle just by calculating the dimension.

Theorem 17 (Frankl-Wilson theorem) *Let $\mathcal{A} \subseteq \mathcal{P}(n)$ be S-intersecting (for $A, B \in \mathcal{A}$ distinct, $|A \cap B| \in S$). Then*

$$|\mathcal{A}| \leq \sum_{i=0}^{|S|}\binom{n}{i}$$

A reduced form of this with the assumption $|A| \notin S$ for $A \in \mathcal{A}$ is immediate from the above, and we don't prove the more general version.

Theorem 18 (Ray-Chaudhuri-Wilson) *Let $\mathcal{A} \subseteq [n]^{(k)}$ be S-intersecting with $S \subseteq \{0, \dots, k-1\}$. Then*

$$|\mathcal{A}| \leq \binom{n}{|S|}.$$

The proof of this requires performing the same polynomial linear independence argument, but also adding an additional $\sum_{i=0}^{|S|-1}\binom{n}{i}$ elements to the spanning set. To do this, we use the functions for $B \in [n]^{(\leq |S|-1)}$,

$$q_B(x) = \left(\sum_{i=0}^{n} x_i - k\right)\prod_{i \in B} x_i.$$

Note that $q_B(\chi_C) = 0$ if $B \not\subseteq C$ or $|C| = k$, and is non-zero otherwise. Thus we can apply the linear combination of this and the previous f_A to the characteristics χ_A for $A \in \mathcal{A}$ to get the coefficients of the f_As equal to zero, and then in increasing order to minimal χ_C. This gives linear independence.

To summarise the collection of theorems:

	$\mathcal{A} \subseteq$	$\lvert A \cap B \rvert \in$	$\lvert A \rvert \notin$	$\lvert \mathcal{A} \rvert \leq$
FW	$\mathcal{P}(n)$	$S \subseteq [n]-1$	$\varnothing$	$\sum_{i=0}^{\lvert S \rvert}\binom{n}{i}$
M-FW	$\mathcal{P}(n)$	$S/p\mathbb{Z}$, $S \subseteq ([p]-1)$	$S/p\mathbb{Z}$	$\sum_{i=0}^{\lvert S \rvert}\binom{n}{i}$
Oddtown	$\mathcal{P}(n)$	$2\mathbb{Z}$	$2\mathbb{Z}$	n
Fisher	$\mathcal{P}(n)$	$\{k\}$	$\varnothing$	n
RCW	$[n]^{(k)}$	$S \subseteq [k]-1$	$\varnothing$	$\binom{n}{\lvert S \rvert}$
G-EKR	$[n]^{(k)}$, $n \geq n_0$	$\{t, \dots, k-1\}$	$\varnothing$	$\binom{n-t}{k-t}$
EKR	$[n]^{(k)}$, $k \leq n/2$	$\{1, \dots, k-1\}$	$\varnothing$	$\binom{n-1}{k-1}$
Trivial	$[n]^{(k)}$, $k > n/2$	$\{1, \dots, k-1\}$	$\varnothing$	$\binom{n}{k}$

From the above, we see that we have a collection of theorems for the general $\mathcal{A}$ case. Note that the Oddtown and Fisher cases are slightly better than special cases, as their argument allows us to remove constants from the vector space (thus decreasing the bound by 1). We also have a few theorems for the k-uniform case, with the most general theorem being RCW. In practice we often just want to take the bound as a polynomial in n, so RCW is almost a total generalisation.

Applications

Theorem 19 (Kahn-Kalai) *Let $k(d)$ be the smallest integer such that every subset of $\mathbb{R}^d$ of diameter 1 can be partitioned into $k(d)$ sets of smaller diameter. Then there is $c > 1$ such that*

$$k(d) \geq c^{\sqrt{d}}$$

for infinitely many d.

Our overarching goal within this proof is to construct, with d an increasing function of p, a set $K \subseteq \mathbb{R}^d$ corresponding to a set of graphs, such that if $L \subseteq K$ has a smaller diameter, by the nature of the identification the graph of L takes up a bounded fraction of the graph of K (which we can then use to bound below the number of such Ls we need to divide K up).

To do this, take $d = \binom{4p}{2}$, and with the natural identification of $\{0,1\}^d$ with subsets of $E(K_{4p})$, write for $A \in [4p]^{(2p)}$, E_A as the complete bipartite graph connecting A and A^c, $\mathcal{F} := \{E_A : A \in [4p]^{(2p)}\}$. We then define our $K \subseteq \mathbb{R}^d$ as the natural identification of $\mathcal{F}$, and note that for $A, B \in [4p]^{(2p)}$, $||v_{E_A} - v_{E_B}||^2 = |E_A| + |E_B| - 2|E_A \cap E_B|$, so thus any $L \subseteq K$ with smaller diameter, for A, B distinct in $\mathcal{A}$ the set of subgraphs corresponding to L, $|E_A \cap E_B|$ cannot be minimal (occurring iff $|A \cap B| = p$), and thus we get our result from the lemma, that

$$|\mathcal{F}|/|\mathcal{A}| \geq \frac{1}{6}\binom{4p}{2p}\Big/\binom{4p}{p-1}$$

and thus covering K requires this number of sets, which one can see is asymptotically exponential in p, which is itself asymptotically equivalent to $\sqrt{d}$.

Lemma 20 *Let p be a prime and $\mathcal{A} \subseteq [4p]^{(2p)}$. If there is no distinct pair $A, B \in \mathcal{A}$ with $|A \cap B| = p$, then $|\mathcal{A}| \leq 4\binom{4p}{p-1}$.*

Take $x \in [4p]$ which occurs in at least half the sets in $\mathcal{A}$ (see existence by a probabilistic argument). Note that for distinct $A, B \in \mathcal{A}$ both containing x, $|A \cap B| \in \{1, \dots, p-1\} \bmod p$, and $|A| = 2p \notin \{1, \dots, p-1\} \bmod p$, so applying M-FW gives

$$\frac{1}{2}|\mathcal{A}| \leq |\{A \in \mathcal{A} : x \in A\}| \leq \sum_{i=0}^{p-1}\binom{4p}{i} \leq 2\binom{4p}{p-1}$$

where the last equality is from rearrangement with the inequality $\binom{4p}{i-1} \leq \binom{4p}{i}/2$, and thus $|\mathcal{A}| \leq 4\binom{4p}{p-1}$ (indeed we can also get $|\mathcal{A}| \leq 3\binom{4p}{p-1}$ by the same method, but this is unnecessary).

VC dimension and traces

The point of VC dimension is to quantify, of a collection of subsets $\mathcal{F}$, what is the largest set of points such that $\mathcal{F}$ is rich enough to encode all binary classifications of them? It can be described as quantifying the complexity or richness of a set collection.

This is a task well-studied by learning theory. In this context, the setup is that we want to understand the *capacity* of a particular model for learning, specifically in the context of binary classification (where we want to learn a function $\widetilde{f} : X \to \{0,1\}$, ideally such that it is equal to some unknown $f : X \to \{0,1\}$). For a set $\mathcal{F} \subseteq \mathcal{P}(X)$, we define the *model* as the collection of candidate classifiers (also known as the set of parametrisations)

$$\mathcal{C} := \{\chi_F : F \in \mathcal{F}\}.$$

The learning model will then, for a particular task instance f, and some collection of training data $D = \{(x, f(x)) : x \in S\}$, attempt to find a solution $\widetilde{f} \in \mathcal{C}$ to minimise $\|f - \widetilde{f}\|$ (with some appropriate norm on $\{0,1\}^X$). Clearly, if $\mathcal{F} = \mathcal{P}(X)$, then such a solution will always exist, although it will take being given all $2^{|X|}$ points to learn it (at which point there are no unknowns to predict!).

The VC dimension of the model $\mathcal{C}$ is then the maximum size of a collection of training data $|D| = |S|$ such that for any $f \in \{0,1\}^X$, there exists $\widetilde{f} \in \mathcal{C}$ such that $\widetilde{f}\big|_S \equiv f\big|_S$. In English: it's the largest collection of data points which the model would always be able to learn to distinguish. If any more points were to be added, then *a priori* the model could fail on the training set not because of limitations of the training procedure, but because it isn't sufficiently complex.

We now give its technical definition, which can be seen to be equivalent to the one above for $\mathcal{C}$.

Definition 3 (Trace) *For $\mathcal{F} \subseteq \mathcal{P}(X)$, $S \subseteq X$, the trace of $\mathcal{F}$ on S is*

$$\mathcal{F}\big|_S := \{F \cap S : F \in \mathcal{F}\}$$

and $\operatorname{tr}_{\mathcal{F}}(S) = \left|\mathcal{F}\big|_S\right|$.

We say that S is shattered by $\mathcal{F}$ if $\mathcal{F}\big|_S = \mathcal{P}(S)$, and define the VC dimension of $\mathcal{F}$ as

$$\max\{|S| : S \subseteq X \text{ is shattered by } \mathcal{F}\}.$$

To give some intuition: as a result of this, we're mainly concerned with being able to divide up a finite set S of elements by the various sets in $\mathcal{F}$. For example, the VC-dimension of the set of half-spaces in $\mathbb{R}^2$ is 3, because it's not possible to draw 4 points on a plane without there being two lines between distinct pairs which intersect, and thus any half plane containing two of these opposite points must also contain their line, and thus the intersection, including a third point.

Theorem 21 (Shauer-Shelah) *If $\mathcal{A} \subseteq \mathcal{P}(n)$ has VC dimension at most d,*

$$|\mathcal{A}| \leq \left|[n]^{(\leq d)}\right|$$

The intuition is that to shatter a set in $[n]^{(d+1)}$, it's sufficient to just have $> \left|[n]^{(\leq d)}\right|$ sets in your system, because at a certain point we have to capture at least one $[n]^{(d+1)}$ set fully.

We may prove this either by induction on $n + d$ or by compression. The compression argument defines $\pi_i : \mathcal{P}(n) \to \mathcal{P}(n)$ for $i \in [n]$ by

$$\pi_i(\mathcal{C}) = \{C \setminus \{i\} : C \in \mathcal{C},\ C \setminus \{i\} \notin \mathcal{C}\} \cup \{C : C \in \mathcal{C},\ C \setminus \{i\} \in \mathcal{C}\}.$$

That is, for each $C \in \mathcal{C}$, we replace it by $C \setminus \{i\}$ if this set doesn't already exist in $\mathcal{C}$. Thus $\pi_i(\mathcal{C}) = \mathcal{C}$ for all $i \in [n]$ iff $\mathcal{C}$ is a down-set, and as π_i otherwise reduces the sum of the cardinalities of $\mathcal{C}$, this procedure terminates.

The key property of this compression is that no set not shattered by $\mathcal{C}$ will be shattered by $\pi_i(\mathcal{C})$, so $\operatorname{VC}(\mathcal{C}) \geq \operatorname{VC}(\pi_i(\mathcal{C}))$. The intuition for this is that highly intersecting set systems may have traces with similarly high levels of intersection, ensuring that it may account for all of the problems in $\mathcal{P}(S)$. As we remove elements, we remove such intersections, and thus may decrease the system's VC dimension.

Strictly, if $\mathcal{C}$ does not shatter S, then there exists $A \subseteq S$ such that for $C \in \mathcal{C}$, $C \cap S \neq A$. Thus the only sets in $\pi_i(\mathcal{C})$ which have a chance are the new ones, and indeed if $C \setminus \{i\} \cap S = A$, then $i \notin A$, so $A \cup \{i\}$ can only be solved by some $C \in \mathcal{C}$. Yet if there existed such a $C \in \mathcal{C}$, then $C \setminus \{i\} \cap S = A$, so $C \setminus \{i\} \notin \mathcal{C}$, so thus $C \notin \pi_i(\mathcal{C})$.

With this as such, we see that the fully compressed system $\mathcal{A}$ is a down-set, and thus has VC dimension equal to the highest cardinality set it contains. If $\mathcal{C}$ has VC dimension $\leq d$, thus the largest element $\mathcal{A}$ can contain has cardinality d, so $|\mathcal{A}| \leq |[n]^{(\leq d)}|$.

Note that as $\mathcal{A}$ shattering S requires that $2^{|S|} = |\mathcal{A}\big|_S| \leq |\mathcal{A}|$, we get $\operatorname{VC}(\mathcal{A}) \leq \log|\mathcal{A}|$. Furthermore, with the (loose) bound of $\sum_{k=0}^{d}\binom{n}{k} \leq n^{d+1}$, we are able to get

$$\frac{\log|\mathcal{A}|}{\log n} - 1 \leq \operatorname{VC}(\mathcal{A}) \leq \log|\mathcal{A}|.$$

and thus $\operatorname{VC}(\mathcal{A}) = \Theta(\log|\mathcal{A}|)$.

Combinatorial Nullstellensatz

Theorem 22 (Combinatorial Nullstellensatz) *Let $\mathbb{F}$ be a field, $f \in \mathbb{F}[x_1, \dots, x_n]$ a polynomial of degree t, with $t_1, \dots, t_n$ such that $\sum_{i=1}^{n} t_i = t$ and $\prod_{i=1}^{n} x_i^{t_i}$ has non-zero coefficient in f.*

If there are sets $S_1, \dots, S_n$ in $\mathbb{F}$ each with $|S_i| \geq t_i + 1$, then there is $\boldsymbol{s} \in \prod_{i=1}^{n} S_i$ such that $f(\boldsymbol{s}) \neq 0$.

The proof is by induction. For $t = 1$, f is linear in an argument, and the statement is immediate. Assuming that there are $S_1, \dots, S_n$ in $\mathbb{F}$ with $|S_i| \geq t_i + 1$, and taking $s_1 \in S_1$,

$$f(x_1, \dots, x_n) = (x_1 - s_1)g(x_1, \dots, x_n) + h(x_2, \dots, x_n)$$

where g is a polynomial of degree $t-1$ with monomial $x_1^{t_1-1}\prod_{i=2}^{n} x_i^{t_i}$, and $h \in \mathbb{F}[x_2, \dots, x_n]$ with degree t. Immediately, for $(s_2, \dots, s_n) \in \prod_{i=2}^{n} S_i$, $f(s_1, s_2, \dots, s_n) = h(s_2, \dots, s_n)$. Furthermore, by the inductive hypothesis there is some $(s_1', \dots, s_n') \in (S_1 \setminus \{s_1\}) \times \prod_{i=2}^{n} S_i$ such that $g(s_1', \dots, s_n') \neq 0$, so

$$\begin{aligned} f(s_1', \dots, s_n') &= (s_1' - s_1)g(s_1', \dots, s_n') + h(s_2', \dots, s_n') \\ &= (s_1' - s_1)g(s_1', \dots, s_n') + f(s_1, s_2', \dots, s_n') \end{aligned}$$

and thus $f(s_1', \dots, s_n') \neq f(s_1, s_2', \dots, s_n')$, so at least one of these is non-zero.

This is naturally an incredibly wide-reaching theorem. The key idea is that given a problem, we want to construct a polynomial with informative zeros, and then take a subset of its domain big enough to guarantee a non-zero value, which indicates that we've exited our initial case.

Theorem 23 *With $H_1, \dots, H_m$ a family of hyperplanes in $\mathbb{R}^n$, and*

$$\left|\{0,1\}^n \cap \bigcup_{i=1}^{m} H_i\right| = 2^n - 1.$$

Then $m \geq n$.

To prove this, construct a polynomial $P(\boldsymbol{x})$ for $\boldsymbol{x} \in \mathbb{R}^n$, which is zero on $\{0,1\}^n$ if $\boldsymbol{x}$ is contained in a hyperplane and non-zero, or equal to zero. This can be easily constructed with degree $n \vee m$, and if $m < n$, the combinatorial nullstellensatz gives us that there is $\boldsymbol{s} \in \{0,1\}^n$ such that $P(\boldsymbol{s}) \neq 0$, so thus there must be a point not contained in a hyperplane.

Theorem 24 (Cauchy-Davenport) *If p is prime and $A, B \subseteq \mathbb{Z}_p$, then*

$$|A + B| \geq \min\{p, |A| + |B| - 1\}.$$

The claim is almost immediate for $|A| + |B| > p$, as we just need to show that $A + B = \mathbb{Z}_p$. In the case $|A| + |B| \leq p$, we want to claim a contradiction if $|A + B| \leq |A| + |B| - 2$, by constructing a polynomial of degree $|A| + |B| - 2$, zero on $A \times B$. Yet combinatorial nullstellensatz should give an element on which the polynomial is nonzero, and this gives us a contradiction.

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